System overview & validated replay

camera_lidar_imu_gps.bag | accepted ~290 s runs

01 / How local motion becomes global position

Sensors
LiDAR / RGB-D / IMU

→

Health & timing
timestamps + confidence

→

Single ESIKF
continuous local state

→

Map alignment
verified map-to-local anchor

Short-term movement is estimated continuously by the ESIKF. Slower geometric matches against the saved map adjust the map-to-local transform, limiting global drift without resetting the filter.

02 / Accepted results | original bag

injected outages: 125–165 s

Mode	Full-run RMSE	Outage RMSE	Outage max	Final diff.
Healthy	**1.384 m**	—	—	**0.462 m**
Camera failed	**1.279 m**	**0.997 m**	**2.975 m**	**1.240 m**
LiDAR failed	**1.631 m**	**2.194 m**	**3.859 m**	**0.484 m**
IMU failed	**2.742 m**	**1.475 m**	**4.673 m**	**0.881 m**

03 / What changes when a sensor fails

Healthy	IMU propagation + LiDAR scan-to-map; camera pose updates gated by confidence.
Camera off	Continue LiDAR / IMU odometry; disable visual measurement path.
LiDAR off	Use IMU attitude + metric RGB-D motion; saved visual/depth map tries to fix global pose.
IMU off	Use RGB-D metric relative pose; LiDAR scan-to-scan relative ICP as fallback.

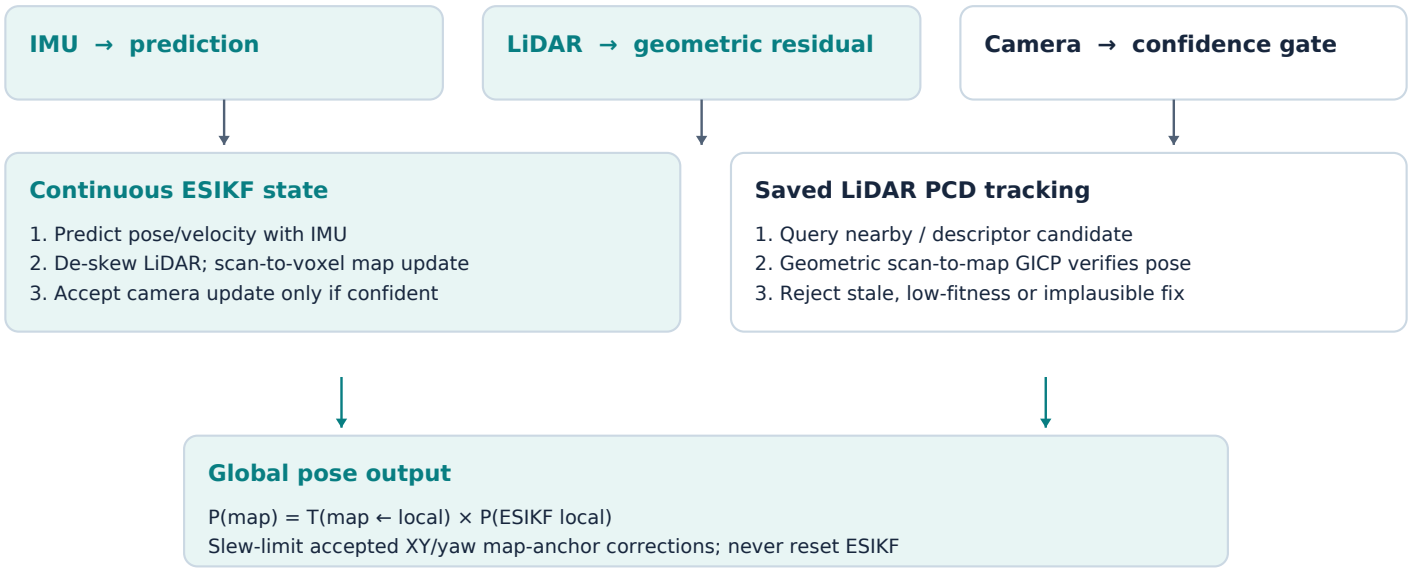
EVALUATION NOTE
Errors are differences against the original-bag trajectory reference, not independent surveyed ground truth. Later October 8 experiments did not replace the full, accepted four-mode 290 s set; therefore all results and plots below use the matched October 6 runs.

Healthy operation & camera dropout

Detailed measurement paths, update checks, and saved-map fusion

01 / Healthy | LiDAR + RGB-D + IMU

FULL_LIVO

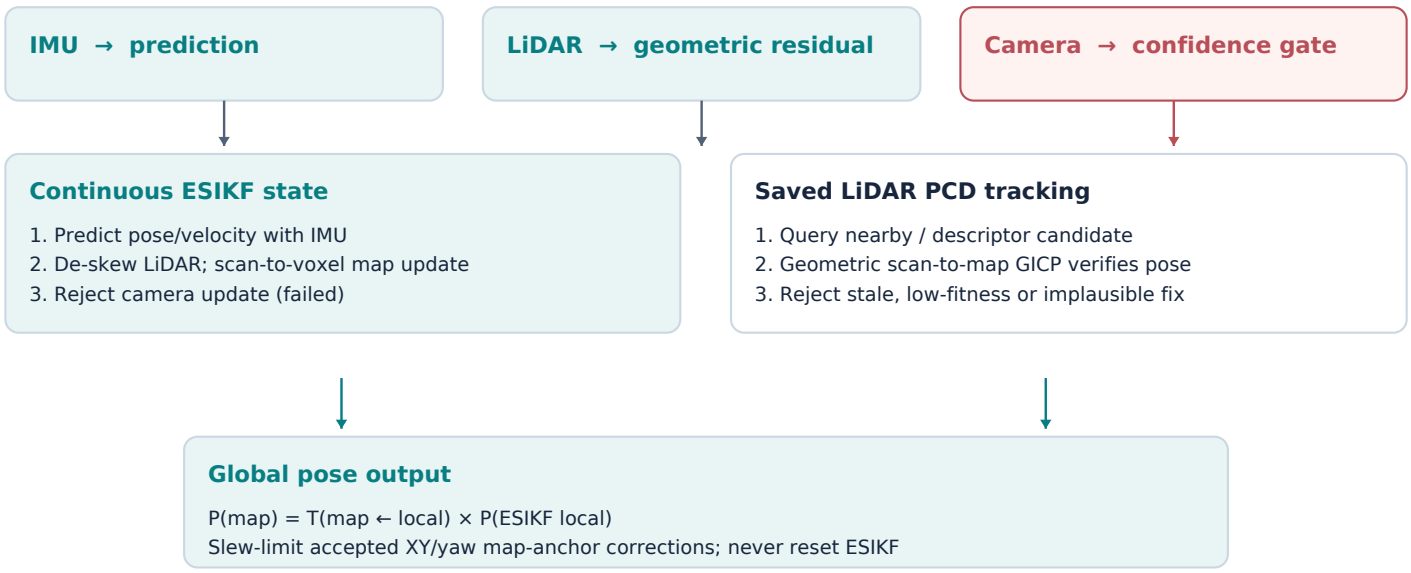


Interpretation

The LiDAR and IMU form a stable motion backbone. Healthy images may enrich the visual map, but uncertain photometric pose updates are gated so they cannot knock the estimator off course.

02 / Camera failure | LiDAR + IMU

LIO_CAMERA_FAILED



Interpretation

Camera-frame timeouts switch to LiDAR-inertial operation. Prediction, scan-to-map optimization, saved-PCD registration and smooth global output all continue in the same estimator.

LiDAR failure | RGB-D + IMU

VI_LIDAR_FAILED / IMU_ONLY_BRIDGE: detailed recovery and confidence gates

03 / Motion when LiDAR scans disappear

Sensor health

LiDAR timeout → failed
RGB-D fresh? confidence gate
Healthy IMU continues

IMU propagation

Gyro → angular update
Accel → velocity/position
Carry continuous ESIKF state

LiDAR map update OFF

No fresh scan residuals
Do not reuse stale map
Keep predicted state

04 / RGB-D relative-motion frontend

Synchronized RGB + depth

Undistort / rectify frames
Track ORB visual features
Associate valid depth

→

Metric motion estimate

3D from depth and intrinsics
Feature motion / PnP
Robust relative SE(3) step

→

ESIKF measurement gate

Reject stale / implausible delta
IMU still controls attitude
Apply bounded translation

05 / Global place retrieval and saved visual/depth map

Place hypotheses

Query stored keyframes
RGB appearance / descriptors
Candidate global poses

→

Depth geometry verification

Live RGB-D → 3D cloud
Register to saved voxel map
Check GICP fitness / overlap

→

Safety and continuity gate

Reject repeated-road aliases
Limit speed / heading jump
Only verified fixes accepted

06 / Single-state and map-frame correction

Motion stays in ESIKF; absolute correction stays in map frame

RGB-D relative increments advance the *same* ESIKF (not a second SLAM trajectory).
Verified depth-map fix adjusts smoothed T(map ← local); output = T × P(ESIKF).
If no unique map fix exists, keep continuous odometry but confidence / drift can degrade.

Practical limitation: visually similar roads can cause false place recognition. The four-mode October 6 replay is retained, but the LiDAR-failure branch still needs robust cross-date validation.

IMU failure | RGB-D + LiDAR

LV_IMU_FAILED: relative-motion source selection without IMU propagation

07 / Motion after IMU messages stop

IMU failure detection

Timestamp timeout; confidence 0
Stop trusting inertial prediction
Preserve last good tilt

RGB-D primary motion

Track RGB features + depth
Solve metric translation + yaw
Relative planar SE(2) increment

LiDAR relative fallback

Scan-to-scan ICP displacement
Use if RGB-D age > 0.75 s
Geometry is backup, not reset

08 / Update the original ESIKF state directly

Same ESIKF state (no replacement)

Position += previous rotation × metric Δp
Yaw += measured Δyaw; keep last valid roll/pitch
Velocity from accepted Δp / Δt

No-IMU propagation / constraints

No fresh IMU acceleration / gyro input
Ground vehicle body-Z constrained
Velocity / yaw rates bounded

09 / Global position correction and recovery

Saved LiDAR geometry

Use existing PCD keyframes
Generate map pose candidate
Match with scan GICP

Verification gate

Fitness, overlap, continuity
Reject large / inconsistent jumps
Maintain map-frame pose

Smooth map anchor

Bound XY, yaw correction rate
Output /pose, /odom, TF
Returning IMU rejoin warmup

Why the no-IMU branch is not a second estimator

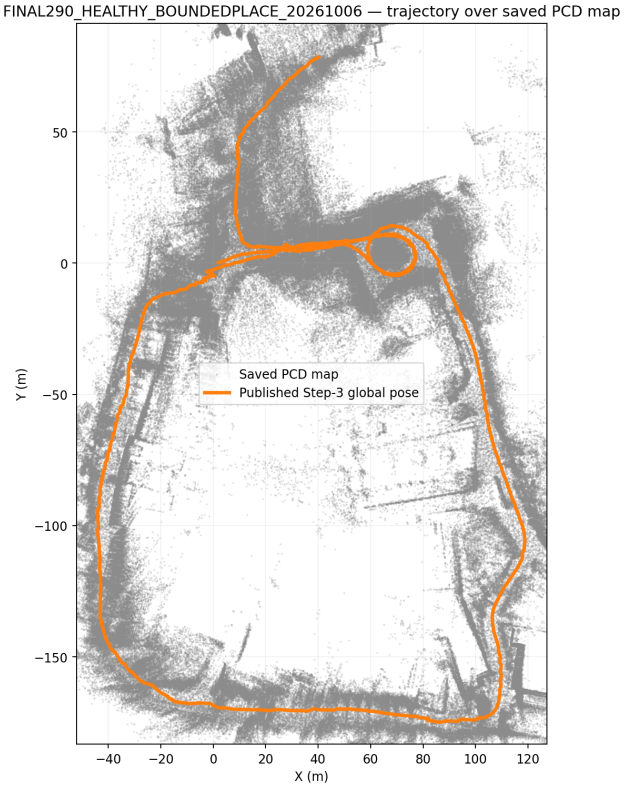
Both RGB-D and LiDAR frontends only provide relative-motion measurements. Their filtered increments update the one surviving FAST-LIVO2 ESIKF state. The saved map supplies an absolute global frame, not an independent parallel trajectory.

The original-bag 40 s IMU outage has a 1.475 m reference-relative RMSE; the whole run is 2.742 m because a large earlier disagreement inflates the full-run score.

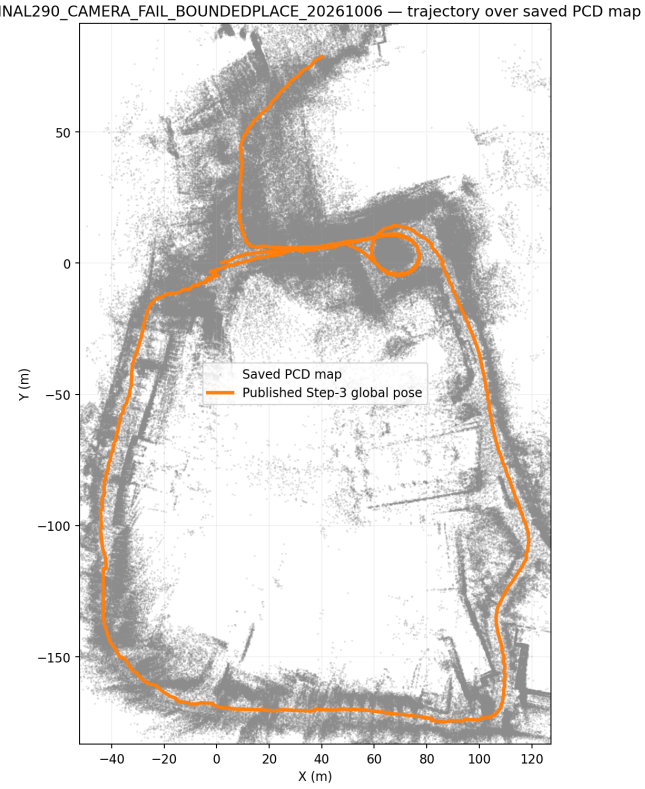
Original saved PCD trajectory PNGs

Embedded directly from plots/07_orange_trajectory_on_pcd_clean.png for the accepted four runs

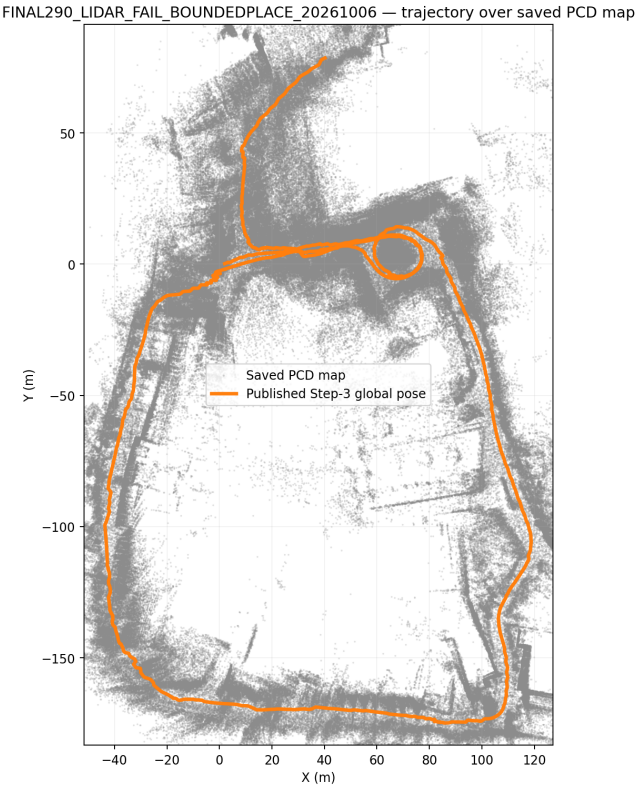
Healthy



Camera failed



LiDAR failed



IMU failed

